GDL Mini-Project*

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1 Introduction

Graph Neural Networks (GNNs) operate by passing messages that aggregate feature information from neighboring nodes across the edges of a graph. A persistent challenge for Message Passing Graph Neural Networks (MPGNNs) is the phenomenon of oversquashing, where information struggles to propagate effectively between distant nodes [8]. This arises either because the underlying graph topology contains bottleneck edges through which a disproportionate amount of information must flow and consequently gets compressed, or because the downstream task requires long-range interactions that the local message passing framework cannot capture.

A growing body of work has proposed graph rewiring as a solution to oversquashing [6, 1] — modifying the graph's edge structure before or during training to improve information flow. These methods fall broadly into two categories. Spatial rewiring methods reduce the commute-time distance between nodes by adding edges within a bounded neighbourhood, while spectral rewiring methods target global connectivity measures such as the Cheeger constant — which quantifies how well different communities within a graph are connected — or the effective resistance between node pairs. Despite their differences, both classes of methods share a critical limitation: they rely exclusively on topological information when deciding which edges to add or remove, entirely ignoring the distribution of node features and the requirements of the downstream task.

In parallel, a separate line of work has reframed MPGNNs within the language of dynamical systems, showing that message passing can be interpreted as the gradient flow of the graph Dirichlet energy [3, 5]. This formulation has been productively used to address oversmoothing — the tendency of deep GNNs to produce indistinguishable node representations — by controlling the balance between attractive and repulsive interactions in the energy functional. More recently, this framework has been extended to characterise oversquashing as an ill-conditioning problem in the energy landscape [4], where bottleneck edges create flat regions through which gradient flow cannot propagate effectively.

This raises a natural question: can the gradient flow perspective on MPGNNs be used to address oversquashing in a task-aware manner, incorporating feature information rather than relying on topology alone?

This work develops a graph rewiring criterion grounded in the Dirichlet energy gradient that identifies bottleneck edges by measuring where gradient flow is most concentrated across the graph. Unlike topology-based methods, our criterion is sensitive to the node feature distribution and the implicit requirements of the downstream task. We demonstrate on controlled synthetic experiments that our method correctly distinguishes between topological bottlenecks that are task-relevant — where inter-community feature variation makes the bottleneck harmful to downstream performance — and those that are not, where topology-only methods rewire wastefully. We further show that our method identifies feature-based bottlenecks in graphs with high curvature where both spectral and spatial rewiring methods have no signal to act on. On real-world node classification benchmarks, we evaluate our approach against topology-based rewiring baselines and show that incorporating feature information into the rewiring criterion leads to improved downstream performance.

*Code available at https://github.com/gdl-final/code

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2 Background

2.1 Oversquashing and Bottleneck Edges

Topping et al. [8] formalise oversquashing through the Jacobian $\partial h_i^{(r)}/\partial x_s$, showing that information from node s is suppressed at node i when few walks exist between them. Edges that lie on most paths between communities — bottleneck edges — cause this suppression by diluting information through the degrees of nodes on both sides. The Balanced Forman curvature characterises these edges locally: positive curvature indicates redundant connectivity through shared triangles or clique membership, near-zero curvature corresponds to grid-like four-cycle structures, and negative curvature identifies tree-like edges where node neighbourhoods are completely disjoint and the edge is the sole connection between two regions. Topping et al. [8] connect this local picture to the global Cheeger constant — measuring the ratio of inter-community to intra-community edges — and, through the discrete Cheeger inequality, to the spectral gap λ_2, the first non-zero eigenvalue of the graph Laplacian. Graphs with topological bottlenecks have small Cheeger constants and consequently small spectral gaps.

FoSR [6] exploits this connection directly, maximising λ_2 by sequentially adding edges via a first-order spectral approximation, while SDRF increases the Balanced Forman curvature of negatively curved edges. Both methods, however, rely entirely on topological information. This reveals a subtle but important limitation: a grid-like graph with near-zero curvature may still suffer from oversquashing if the feature distribution creates task-relevant gradients across certain edges, while a tree-like structure with strongly negative curvature may not require rewiring at all if the feature signal does not cross the topological bottleneck — rendering topology-only rewiring either insufficient or wasteful for the downstream task.

Topping et al. [8] further identify that diffusion-based rewiring methods such as Personalised PageRank — which incorporate feature information through a global random walk on the graph — nonetheless fail to address oversquashing because they preferentially add edges at short diffusion distance, reinforcing intra-community connections rather than the inter-community edges that increase the Cheeger constant and reduce bottleneck severity. Our method avoids this pitfall: by grounding the rewiring criterion in the Dirichlet energy gradient, we identify edges where gradient flow is concentrated across community boundaries, incorporating the feature signal while remaining sensitive to the inter-community structure that governs the Cheeger constant and spectral gap. Barbero et al. [1] identify a complementary tension between spatial rewiring methods, which respect locality but sacrifice sparsity, and spectral methods, which preserve sparsity but violate locality — addressing this through a multi-relational framework that constrains added edges by shortest-walk distance.

2.2 Gradient Flow Formulations of GNNs

Chamberlain et al. [3] establish that GNN architectures can be interpreted as discretised diffusion PDEs on graphs, connecting the layer index to the time discretisation of the heat equation. Di Giovanni et al. [5] extend this by characterising message passing as the gradient flow of a parameterised energy that generalises the Dirichlet energy. The key contribution is showing that the channel-mixing matrix W governs two competing dynamics: positive eigenvalues produce attractive interactions that encourage feature smoothness, driving the system toward oversmoothing, while negative eigenvalues produce repulsive interactions that amplify high-frequency components, leading to oversharpening. This energy-based view provides the backbone for our work.

The connection to oversquashing arises directly from the spectral perspective: graphs with bottleneck edges have a small Fiedler value λ_2, which causes the Hessian of the energy to be ill-conditioned and gradient flow to stagnate in the flat directions corresponding to cross-bottleneck information flow. Eliasof et al. [4] address this dynamically through a learned Lyapunov energy and a tangential component that navigates flat energy regions without modifying the graph. The present work takes the complementary structural approach — using the energy gradient itself to identify and rewire the edges responsible for this ill-conditioning.

3 Methodology

3.1 GRAFF: Message Passing as Gradient Flow

We build on the gradient flow formulation of Di Giovanni et al., which characterises message passing as the evolution of node features under the gradient of a parameterised energy functional. A GNN layer of the form

$$F(t+1) = F(t) + AF(t)W \tag{1}$$

where A is the normalised adjacency matrix and W is a symmetric channel-mixing matrix, can be interpreted as a forward Euler discretisation of the gradient flow of a parameterised Dirichlet energy.

The resulting dynamics can be written as the following spectral decomposition:

$$\mathrm{vec}(F(m\tau)) = \sum_{r=0}^{d-1}\sum_{\ell=0}^{n-1} \left(1 + \tau\mu_r(1-\lambda_\ell)\right)^m c_{r,\ell}(0)\, \psi_r \otimes \phi_\ell \tag{2}$$

where μ_r are the eigenvalues of W, λ_ℓ are the eigenvalues of the normalised Laplacian L, ϕ_ℓ are the Laplacian eigenvectors, and $c_{r,\ell}(0)$ are the initial Fourier coefficients of the feature channels.

Positive eigenvalues $\mu_r > 0$ amplify low-frequency modes (small λ_ℓ), driving the system toward oversmoothing. Negative eigenvalues $\mu_r < 0$ amplify high-frequency modes (large λ_ℓ), producing oversharpening. The evolution of features is therefore governed jointly by the spectra of W and L. Hence, our approach is motivated by building a spectral explanation of oversquashing within the gradient flow framework of GRAFF.

3.2 TANGO: Dynamics in Ill-Conditioned Energy Landscapes

Eliasof et al. (TANGO) address the same ill-conditioning phenomenon from a dynamical systems perspective. Rather than modifying the graph structure, they modify feature evolution by decomposing updates into an energy descent component and a tangential component orthogonal to the energy gradient:

$$\frac{dH}{dt} = -\alpha_G(H)\nabla_H V_G(H) + \beta_G(H)\, T_{V_G}(H) \tag{3}$$

where the tangential component is defined by projecting a learned update direction $M(H)$ onto the subspace orthogonal to the gradient:

$$T_{V_G}(H) = M(H) - \langle M(H), \hat{\nabla}_H V_G(H)\rangle\, \nabla_H V_G(H). \tag{4}$$

This construction ensures $\langle T_{V_G}(H), \nabla_H V_G(H)\rangle = 0$, so the tangential component moves features along level sets of the energy without changing its value. The gradient component decreases energy, while the tangential component enables continued feature evolution in flat regions of the energy landscape.

These flat regions arise in graphs with bottlenecks, where the spectral gap λ_2 is small and the energy landscape is poorly conditioned. In such regions, gradient descent alone is ineffective, while the tangential component allows traversal of level sets until informative descent directions are found. Our approach builds on the formulation of oversquashing of gradient flow as targets the structural cause directly by modifying the graph topology to increase λ_2 and improve conditioning prior to feature propagation.

3.3 Spectral Gap, Cheeger Constant, and Bottlenecks

The spectral gap λ_2—the first non-zero eigenvalue of the normalised Laplacian—governs the rate of information mixing under diffusion. By Cheeger's inequality,

$$\frac{\lambda_2}{2} \leq h(G) \leq \sqrt{2\lambda_2}, \tag{5}$$

where $h(G)$ is the Cheeger constant. Graphs with bottlenecks have small Cheeger constant, as the number of edges between communities is low, and correspondingly small λ_2, leading to slow information propagation and ill-conditioned gradient flow dynamics.

3.4 Spectral Interpretation of the Gradient Signal

Let $X_r \in \mathbb{R}^n$ denote a feature channel. Since the graph Laplacian L is real symmetric, the spectral theorem gives a complete orthonormal eigenbasis $\{\phi_\ell\}$ with $L\phi_\ell = \lambda_\ell \phi_\ell$. Any feature channel can be decomposed as

$$X_r = \sum_{\ell=0}^{n-1} c_{r,\ell}\phi_\ell, \quad c_{r,\ell} = \phi_\ell^\top X_r. \tag{6}$$

One step of gradient flow diffusion with step size η gives

$$X_{\text{smooth}} = X - \eta L X, \tag{7}$$

and applying the decomposition channel-wise yields

$$(X_{\text{smooth},r})_i = \sum_\ell (1 - \eta\lambda_\ell)\, c_{r,\ell}\, \phi_\ell(i). \tag{8}$$

The diffusion step attenuates each frequency component by $(1 - \eta\lambda_\ell)$: low-frequency modes (λ_ℓ small) are preserved, while high-frequency modes are suppressed. The smoothed feature difference across an edge (u, v) is therefore

$$\|X_{\text{smooth}}[u] - X_{\text{smooth}}[v]\|^2 = \sum_r \left(\sum_\ell (1 - \eta\lambda_\ell)\, c_{r,\ell}\, (\phi_\ell(u) - \phi_\ell(v)) \right)^2. \tag{9}$$

The Fiedler vector ϕ_2, corresponding to λ_2, encodes the primary graph partition (Fiedler, 1973; Mohar, 1991). For a bottleneck edge (u, v) separating communities C_A and C_B, ϕ_2 takes approximately constant but opposite signs on each side, giving $|\phi_2(u) - \phi_2(v)|$ large. The corresponding Fiedler contribution to the smoothed feature difference is

$$\|X_{\text{smooth}}[u] - X_{\text{smooth}}[v]\|^2 \approx (1 - \eta\lambda_2)^2\, (\phi_2(u) - \phi_2(v))^2\, \|C_2\|^2, \tag{10}$$

where $C_2 = (c_{1,2}, \ldots, c_{d,2})$ collects the Fiedler projections across all feature channels. The factor $(1 - \eta\lambda_2)$ is close to 1 for small λ_2 (the bottleneck regime), so the Fiedler term is not attenuated by the diffusion step — the bottleneck signal is preserved.

Higher eigenvectors ϕ_ℓ, $\ell \geq 3$, correspond to increasingly oscillatory and spatially localised modes. Their contribution to the smoothed feature difference is limited by three factors. First, their spatial locality implies $\phi_\ell(u)$ and $\phi_\ell(v)$ are not systematically separated at bottleneck edges. Second, their oscillatory nature causes sign variation across edges and feature channels, so contributions cancel when aggregated. Third, the diffusion step attenuates higher frequencies more strongly — the factor $(1 - \eta\lambda_\ell)$ is smaller for larger λ_ℓ, further suppressing these contributions. The dominant, stable contribution to the smoothed feature difference at bottleneck edges therefore arises from the Fiedler component.

3.5 Feature Dependence and Task Awareness

The quantity $\|C_2\|^2 = \sum_r c_{r,2}^2$ measures how much the node features project onto the Fiedler direction — equivalently, how much the feature distribution differs systematically between the two communities separated by the bottleneck. When features differ across communities, $\|C_2\|$ is large and the smoothed feature difference at the bottleneck edge is large, producing a high score. When features are uninformative and uncorrelated with the community structure, $c_{r,2} \approx 0$ for all channels and the Fiedler term vanishes:

$$\|X_{\text{smooth}}[u] - X_{\text{smooth}}[v]\|^2 \approx 0 \tag{11}$$

regardless of how severe the topological bottleneck is — that is, regardless of how large $|\phi_2(u) - \phi_2(v)|$ is. The score therefore carries no signal about the topological bottleneck when features are task-irrelevant, and the method correctly abstains from rewiring.

This distinguishes our method from topology-only approaches such as FoSR, whose score $x_2(u) \cdot x_2(v)$ depends only on the Fiedler vector entries and is independent of $\|C_2\|$. FoSR identifies the same topological bottleneck regardless of the feature distribution, rewiring edges that may carry no task-relevant signal. Our score recovers the FoSR criterion in the limit $\|C_2\| \to \infty$ (features perfectly aligned with the community partition) and degrades to zero in the limit $\|C_2\| \to 0$ (uninformative features), interpolating task-awareness between these extremes.

3.6 Gradient-Based Rewiring

Given a graph $G = (V, E)$ with node features $X \in \mathbb{R}^{n \times d}$, we compute the Dirichlet energy gradient LX and apply one step of gradient flow diffusion with adaptive step size:

$$X_{\text{smooth}} = X - \eta LX, \quad \eta = \frac{0.9}{d_{\max} + \epsilon} \tag{12}$$

where $d_{\max}$ is the maximum node degree. The step size satisfies the stability condition $\eta < 1/d_{\max}$ for the unnormalised Laplacian, ensuring the diffusion does not overshoot.

The bottleneck score for each edge $(u, v) \in E$ is defined as

$$\text{score}(u, v) = \frac{\|X_{\text{smooth}}[u] - X_{\text{smooth}}[v]\|}{\|X_{\text{smooth}}[u]\| + \|X_{\text{smooth}}[v]\| + \epsilon}. \tag{13}$$

This score measures how much two endpoints diverge in representation space after one gradient flow step, normalised by their magnitude. As shown in Section 3.4, it is dominated by the Fiedler component when features carry inter-community signal and vanishes when $\|C_2\| \approx 0$. The normalisation by endpoint magnitudes addresses a practical instability of the unnormalised gradient difference $\|(LX)_u - (LX)_v\|^2$: bridge nodes accumulate large gradient magnitude from aggregating across community boundaries, which inflates the score of all their incident edges including intra-community ones. The normalised score resolves this — an intra-community edge incident to a bridge node has a large denominator but small numerator, since both endpoints converge toward the same community centroid after diffusion, and is correctly ranked lower than the bridge edge itself.

For a selected bottleneck edge (u, v), candidate supporting edges are drawn from $\mathcal{N}(u) \times \mathcal{N}(v)$. For each candidate (a, b), we approximate the effect of insertion using a first-order update:

$$LX_{\text{approx}}[a] = LX_{\text{smooth}}[a] - \frac{1}{d_a + 1} \left(LX_{\text{smooth}}[a] - LX_{\text{smooth}}[b]\right) \tag{14}$$

$$LX_{\text{approx}}[b] = LX_{\text{smooth}}[b] + \frac{1}{d_b + 1} \left(LX_{\text{smooth}}[a] - LX_{\text{smooth}}[b]\right) \tag{15}$$

with X_{approx} updated accordingly. This avoids recomputing the full Laplacian for each candidate, analogous to FoSR's first-order spectral gap approximation. The candidate that maximally reduces the mean bottleneck score over the local neighbourhood $\mathcal{N}(u) \cup \mathcal{N}(v)$ is selected. If all candidate pairs already have edges between them, the algorithm advances to the next highest-scoring bottleneck edge. The full procedure can be found in provided code.

4 Empirical Setup

- **Graph 1 — Chain of cliques with informative features.** A chain of five fully-connected cliques connected by bridge edges of varying severity (11 bridge edges total), and a two-clique barbell connected by a single bridge edge. Node features are drawn from distinct Gaussian distributions per clique, with σ_{signal} controlling inter-clique variation and σ_{noise} controlling intra-clique noise. The SNR ratio $\sigma_{\text{signal}}/\sigma_{\text{noise}}$ directly corresponds to the theoretical condition $\|C_2\| > 0$ from our spectral decomposition: when $\sigma_{\text{signal}} > \sigma_{\text{noise}}$, features project onto the Fiedler vector and our criterion correctly identifies bridge edges. This tests the core detection case where topology and feature signal are aligned.
- **Graph 2 — Chain of cliques with uninformative features.** Same topology as Graph 1 but all cliques share an identical feature distribution — inter-clique feature variation is zero by construction, corresponding to $C_2 \approx 0$ in our theoretical framework. This tests the task-awareness property: our criterion should assign near-uniform scores (separation ≈ 1, precision $\approx$ random) while FoSR should consistently identify the topological bridge edges regardless of feature content, demonstrating wasteful rewiring when features provide no signal across the bottleneck.
- **Graph 3 — Dense random graph with feature gradient.** An Erdős–Rényi graph with edge probability $p = 0.3$, giving high spectral gap and no topological bottleneck — the graph has positive curvature throughout and FoSR has no spectral signal to act on. Two latent node groups have systematically different feature distributions, with σ_{signal} and σ_{noise}

controlling the clarity of the feature gradient across the group boundary. The primary metric is lift over the random baseline — how much above chance each method identifies the cross-boundary edges in its top-k. This tests the case from our theoretical analysis where the Cheeger constant is large but the feature gradient creates a task-relevant bottleneck invisible to topology-only methods.

We evaluate downstream accuracy using GRAFF [5] as our backbone, comparing three conditions: no rewiring, FoSR preprocessing (using the implementation provided by [6]), and our gradient-based rewiring, each applied once before training.

Datasets. We evaluate on Cornell and Texas from the WebKB collection [7], which are heterophilic webpage classification graphs with 5 classes. We select these datasets because their small size (183 nodes, $\sim$300 edges) means our rewiring budget of $k \in \{10, 20, 30\}$ represents 3–10% of total edges, making rewiring proportionally significant. Both datasets are included in GRAFF's original evaluation [5], providing directly comparable baselines. We note that our feature SNR diagnostic gives values below 0.1 for both datasets, placing them in the low-SNR regime where our synthetic analysis predicts neither method should show a strong advantage — we include them to verify that our method does not actively harm performance when features are uninformative, consistent with our Graph 2 results. We follow GRAFF's experimental protocol exactly, using the same fixed dataset splits, model architecture, and training configuration as reported in [5], with results averaged over 10 random seeds.

5 Results

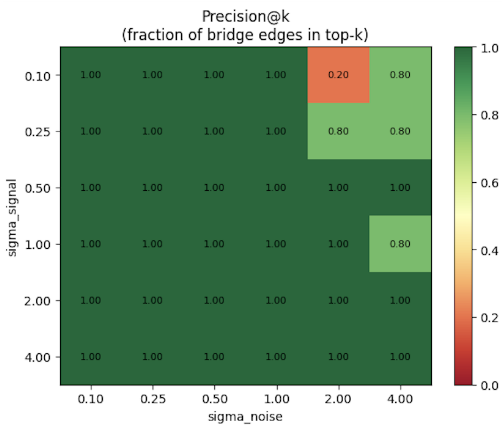
We present results across three synthetic graph families. Table 1 reports precision@k at SNR=1 ($\sigma_{\text{signal}} = \sigma_{\text{noise}} = 1$), representing the boundary condition where inter-clique signal exactly matches intra-clique noise.

Graph 1 and Graph 2 — Chain of Cliques. On the two-clique barbell both methods achieve precision 1.00 under informative features, consistent with the theory: when the graph has exactly two communities, the Fiedler vector cleanly separates them and both the topological criterion (FoSR) and the feature-aware criterion (GBR) identify the same bridge edge. On the five-clique chain the difference is stark. FoSR achieves precision 0.17 regardless of whether features are informative or uninformative — the identical performance across both columns of Table 1 directly confirms its feature-agnostic nature. This is expected: FoSR's score depends entirely on the second eigenvector of the adjacency, which on a five-clique chain captures the global spectral cut between the two outermost cliques rather than the individual bridges between adjacent pairs. GBR achieves precision 1.00 under informative features by scoring each edge locally from the gradient flow, identifying each community boundary independently without relying on a single global partition.

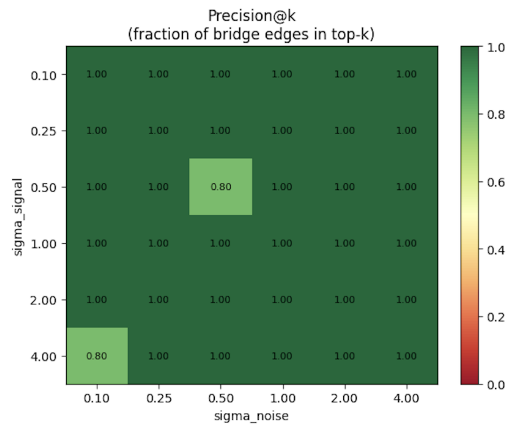
Under uninformative features GBR's precision drops to 0.22 — near the random baseline. This is the correct behaviour predicted by our framework: when $C_2 \approx 0$ the score degenerates to noise-dominated variation and the bridge edges are indistinguishable from intra-clique edges. Crucially this is not a failure — it reflects the correct conclusion that when features carry no inter-community signal, rewiring the topological bottleneck provides no task benefit. FoSR rewires regardless, which increases λ_2 but does not help the downstream task.

Figure 1 shows the full SNR analysis. GBR's precision transitions sharply from near-zero to near-perfect around SNR ≈ 1, corresponding to the theoretical condition $\|C_2\|^2 > 0$. FoSR's precision is flat across the entire SNR grid, confirming feature-agnostic behaviour.

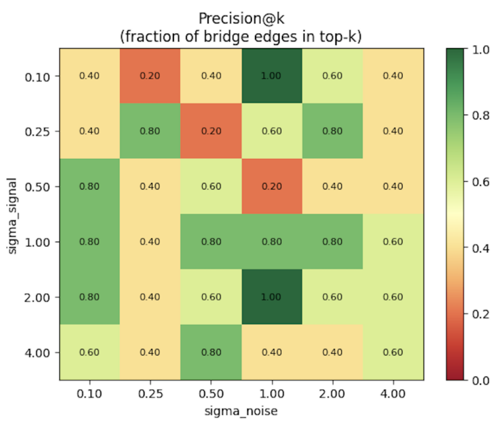
Graph 3 — Dense Random Graph with Feature Gradient. FoSR achieves lift at or below 1.0 across all SNR values, confirming that it has no signal to act on when the graph is topologically well-connected. GBR achieves lift consistently above 1.0 when SNR ≥ 1, identifying cross-boundary edges at approximately twice the random baseline. At low SNR, GBR's lift approaches 1.0 — the correct behaviour when the feature signal is not distinguishable from noise.



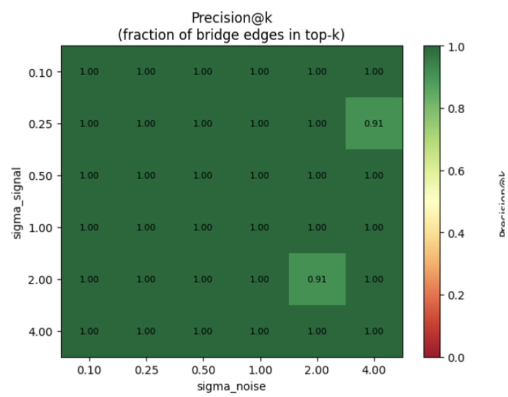
(a) GBR — Informative features



(b) FoSR — Informative features



(c) GBR — Uninformative features

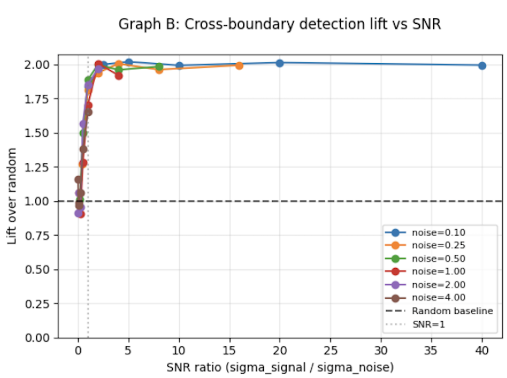


(d) FoSR — Uninformative features

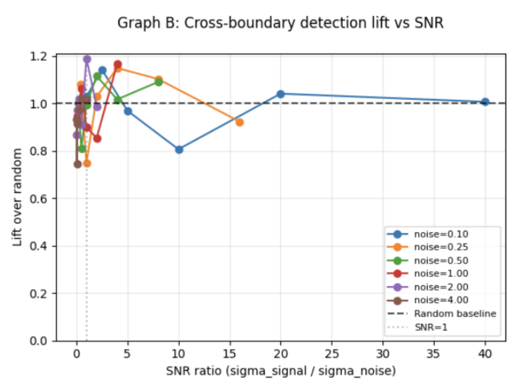
Figure 1: Synthetic comparison — two-clique setting.

Table 1: Precision@k at $\sigma_{\text{signal}} = \sigma_{\text{noise}} = 1$.

Method	Informative features	Uninformative features
GBR	**1.00**	0.22
FoSR	0.17	**0.52**



(a) GBR — Feature bottleneck



(b) FoSR — Feature bottleneck

Figure 2: Dense graph with no topological bottleneck. GBR identifies the feature-induced bottleneck, while FoSR does not.

Benchmark Experiments.

Tables 2 and 3 report results on Texas and Cornell. On Texas, GBR achieves 79.73 ± 4.87 at $k = 20$ compared to 75.68 ± 8.55 for no rewiring and 72.97 ± 8.63 for FoSR — a positive direction across both evaluation metrics. On Cornell, results across all three methods are within noise at all k values.

However, standard deviations of 7–10% across all methods indicate high variance, consistent with the small size of WebKB datasets. Both datasets lie in the low-SNR regime, where inter-class feature variation is dominated by noise. In this setting, neither method is expected to show strong gains, and GBR correctly avoids degrading performance.

Table 2: Texas results (mean ± std).

k	Metric	No rewiring	FoSR	Ours
10	Test @ best val	75.68 ± 8.55	71.89 ± 8.39	75.14 ± 4.95
	Dirichlet energy	40.54 ± 6.23	37.77 ± 7.19	31.99 ± 5.70
	Last-10 avg	72.97 ± 8.80	74.05 ± 10.48	75.41 ± 7.20
20	Test @ best val	75.68 ± 8.55	72.97 ± 8.63	79.73 ± 4.87
	Dirichlet energy	40.54 ± 6.23	39.32 ± 6.45	33.45 ± 5.10
	Last-10 avg	72.97 ± 8.80	74.32 ± 7.38	78.38 ± 6.16

Table 3: Cornell results (mean ± std).

k	Metric	No rewiring	FoSR	Ours
10	Test @ best val	50.81 ± 8.53	50.81 ± 7.72	50.81 ± 9.58
	Dirichlet energy	12.54 ± 3.60	12.38 ± 2.96	14.40 ± 3.73
20	Test @ best val	50.81 ± 8.53	49.46 ± 7.26	49.73 ± 7.17
	Dirichlet energy	12.54 ± 3.60	16.19 ± 6.38	14.66 ± 5.09
50	Test @ best val	50.81 ± 8.53	48.11 ± 8.00	53.78 ± 7.49
	Dirichlet energy	12.54 ± 3.60	15.28 ± 4.20	17.01 ± 4.67

6 Discussion & Concluding Remarks

This work develops a gradient-flow-based graph rewiring criterion that identifies bottleneck edges using both topological structure and node feature information, consequently addressing oversquashing in a task-aware manner. Our synthetic experiments demonstrate that an energy-based rewiring criterion can address both topology-based and feature-based oversquashing — correctly identifying bridge edges when features carry inter-community signal, correctly abstaining when they do not, and finding feature-gradient bottlenecks invisible to topology-only methods. The benchmark results on Cornell and Texas, while inconclusive due to high variance, confirm that the method does not harm performance in the low-SNR regime. The preserved Dirichlet energy across all k values provides a secondary confirmation: because our criterion adds edges where the feature gradient concentrates at community boundaries rather than within communities, the added connections link nodes with different features and do not encourage feature homogenisation during training.

This places our method in an interesting position relative to the LASER framework of Barbero et al. [1], which requires rewiring methods to simultaneously reduce oversquashing, preserve sparsity, and preserve locality. Like FoSR [6], our method does not strictly respect graph-distance locality — added edges may connect topologically distant nodes. However, it enforces a weaker but arguably more meaningful notion: task-aware locality. Edges are only added where the feature gradient indicates inter-community signal exists and the downstream task can plausibly benefit, rather than wherever the topological structure alone suggests a bottleneck. In this sense, the method is local with respect to the task rather than the topology, a distinction that may be more relevant for heterophilic and long-range interaction graphs where topological proximity is a poor proxy for task relevance.

However, some limitations include that while we provide a theoretical connection between our scoring criterion and the Fiedler vector and Cheeger constant — showing that high-scoring edges correspond to edges where the Fiedler vector takes opposite signs in the presence of inter-community feature signal — this falls short of a formal proof that edge additions increase the spectral gap. A rigorous guarantee analogous to FoSR's Theorem 4 [6] remains future work. More critically, demonstrating feature-awareness on synthetic graphs with controlled SNR is not sufficient to establish task-awareness in the broader sense. Task-awareness requires showing that the added edges improve performance on downstream tasks that require long-range information flow — a condition not met by the small heterophilic benchmarks evaluated here, where the classification task depends primarily on local structural patterns rather than cross-community propagation. This aligns with the oversquashing analysis of Topping et al. [8], which identifies long-range dependency as the critical regime where bottlenecks are harmful.

The natural extension is to graph classification datasets where labels depend on global properties: TUDatasets, Peptides-func, and the LRGB suite all present this requirement. Most relevant are neuroimaging datasets such as NeuroGraph [2], where nodes represent brain regions and edges encode functional connectivity. These graphs have strong intra-community feature similarity — regions within the same functional network have correlated activation patterns — and systematic inter-community variation, placing them in the high-SNR regime where our criterion's advantage over topology-only methods is predicted. The biological motivation is also direct: a topology-only method rewiring based on spectral structure has no basis for distinguishing which functional boundaries are relevant to a given cognitive task, while a feature-aware criterion can in principle identify these boundaries from the activation patterns themselves. Evaluation on such datasets, which was beyond the scope of this study due to the time required to extend GRAFF [5] to graph classification, represents the most important direction for establishing task-awareness beyond the synthetic setting.

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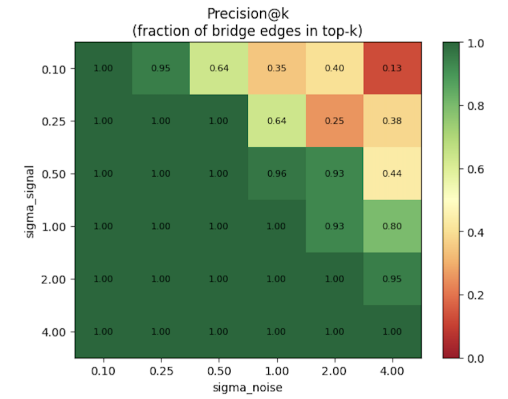
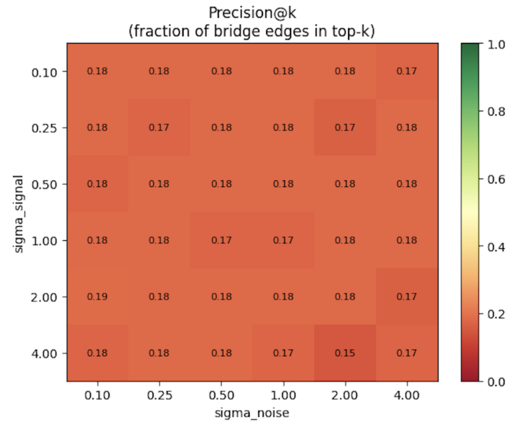
7 Appendix

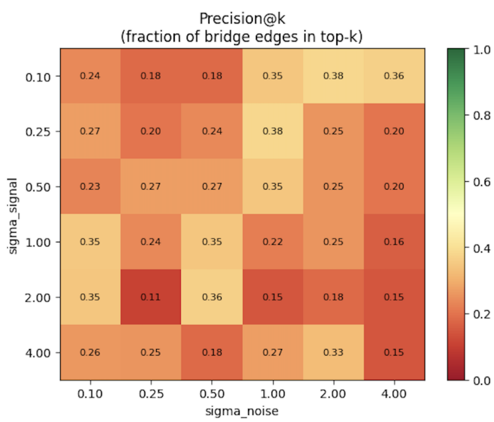
7.1 More Figures



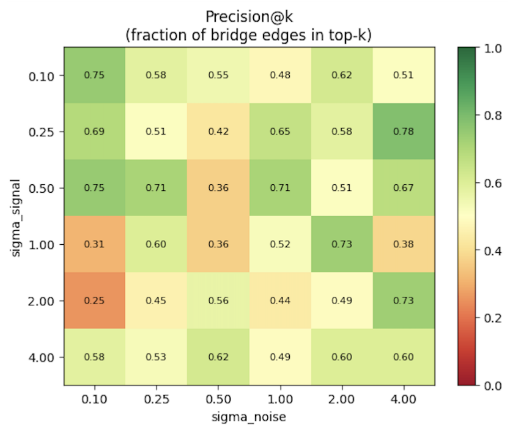
(a) GBR — Informative features



(b) FoSR — Informative features

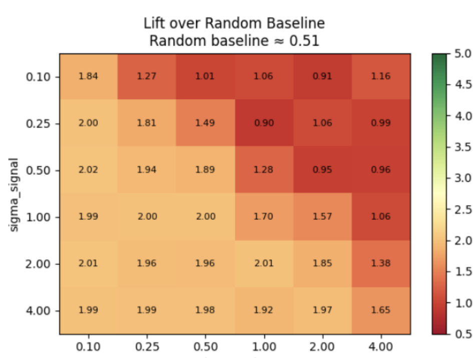


(c) GBR — Uninformative features

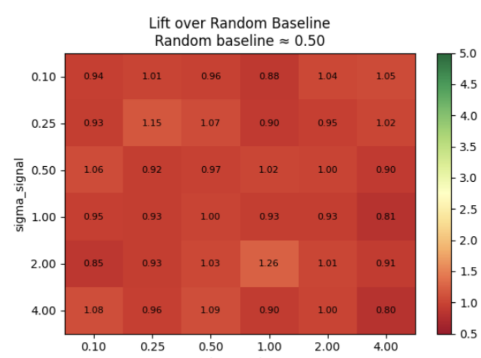


(d) FoSR — Uninformative features

Figure 3: Synthetic comparison - Multi Clique setting. Top: informative features. Bottom: uninformative features. Left: Gradient-based rewiring. Right: FoSR.



(a) GBR — Feature Bottleneck



(b) FoSR — Feature Bottleneck

Figure 4: Comparison on a dense graph with no topological bottleneck.